Effectiveness, Generalizability, and Explainability of MoE-Enhanced Dense Retrievers*

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Abstract

Dense Retrieval Models (DRMs) capture semantic relationships beyond lexical matching and perform well across a range of IR tasks, yet adapting them to new tasks and domains is costly, and since they operate as black boxes, their ranking decisions are hard to interpret. This extended abstract connects a line of work that addresses both issues through a proposed lightweight architecture for enhancing DRMs. A single Mixture-of-Experts (MoE) layer is placed atop a bi-encoder, enriching and contextualizing query and document embeddings before similarity estimation. We investigate two expert routing strategies that guide embedding contextualization, one learned in an unsupervised manner and one weakly supervised by an external signal. We further support our studies' outcomes by reporting new zero-shot results on 13 BEIR collections with five different backbones, to show how the proposed MoE-enhanced DRMs transfer across tasks and domains. A representation-level analysis of the MoE-induced embedding spaces further shows that expert routing draws relevant documents relatively closer to their queries as the space expands, and that experts specialize along linguistic rather than topical properties. Overall, the proposed MoE strategy offers an inexpensive and interpretable way to improve the generalizability of DRMs, independently of the underlying model architecture.

Keywords

Dense Retrieval, Mixture-of-Experts, Explainability in IR

1. Introduction

Dense Retrieval Models (DRMs) encode both queries and documents into a shared embedding space and rank documents by their similarity to the query, capturing semantic relationships that mere lexical matching models might miss [4]. However, they tend to generalize poorly without further adaptation [5]. Strong in-domain performance usually relies on task-specific fine-tuning, which is expensive to repeat for every new collection and impractical in low-resource settings. Additionally, DRMs behave as black boxes, offering little insight into document ranking decisions. Mixture-of-Experts (MoE) [6] offers a possible mitigation to the generalization problem. By routing each input through specialized sub-networks, MoE enriches representations while keeping additional costs low. Prior work in IR has typically replaced the feed-forward layer inside every Transformer layer with a set of experts [7, 8, 9], which benefits the underlying model but increases the parameter count substantially. The works presented here take a different stance by placing a single MoE layer atop the final Transformer layer of the underlying bi-encoder, which is applied to both the query and document embeddings. The experts refine the embedding produced by the backbone, and a gating mechanism weights their outputs, which are then aggregated into a single representation used for the final similarity estimation. Contextualization therefore happens at the full embedding level, rather than through token-level query/document interactions, costing only a small number of additional parameters. We instantiate the proposed MoE design with two different gating mechanisms. SB-MoE [1] learns its expert routing strategy directly from the retrieval signal that trains the whole DRM for the downstream IR task, allowing for dynamic expert selection and specialization. DenseC3 [2] instead guides expert routing with external linguistic information, so that query and document representations sharing similar

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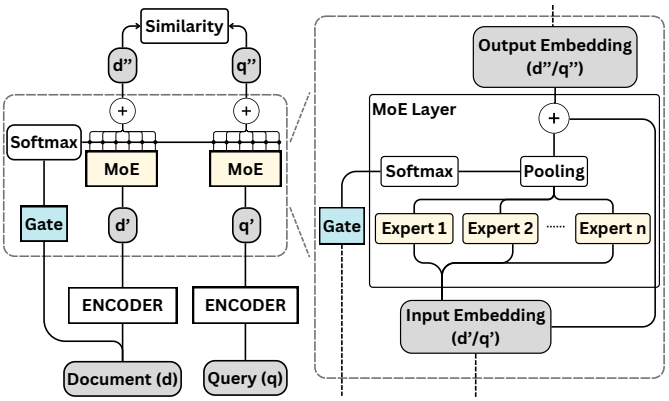


Figure 1: Overview of the proposed MoE-enhanced DRM architecture.

characteristics according to that signal are aligned in the same region of the induced embedding space. The two approaches therefore instantiate an unsupervised and a weakly supervised expert routing strategy, respectively, and comparing them isolates the contribution of each gating mechanism to the retrieval effectiveness of the underlying DRM on the downstream IR tasks.

This extended abstract expands these studies with additional results and unifies their findings to derive broader insights about MoE in dense retrieval. We report a unified zero-shot evaluation in which each model is trained once on MSMARCO and tested on 13 BEIR collections across five underlying DRMs. Additionally, we summarize a representation-level analysis of the resulting embedding spaces [3] to address the interpretability of the proposed MoE-enhanced DRMs by inspecting the induced retrieval spaces. Our findings show that the proposed MoE-enhanced retrievers outperform standard fine-tuning approaches across a majority of collections and backbones. Finally, the MoE-induced embedding spaces are structured, as expert routing pulls relevant documents into relatively tighter query neighborhoods, with experts partitioning the expanding space based on linguistic rather than topical properties.

2. Background & Related Work

Neural IR models differ in how they conduct embedding contextualization. Cross-encoders leverage early token-level query/document interactions, achieving strong effectiveness at a cost that confines them to re-ranking [10]. Bi-encoders drop all interactions, allowing document embeddings to be computed offline and used directly for first-stage retrieval [11]. Late-interaction models retain independent encoding while deferring token-level interactions to retrieval time [12]. Across these families, contextualization enriches query and document representations before relevance estimation, either from their interactions [13] or externally [14]. MoE [15] provides a mechanism for both, with a set of expert sub-networks that process the same input, while a gating mechanism selects the active experts that determine the final representation [16]. In IR, MoE has mostly been integrated inside the DRM, replacing the feed-forward layer of each Transformer layer with multiple experts for passage retrieval [7, 9] and question answering [8]. These designs benefit the underlying model but multiply its parameters and confine embedding contextualization to the token level. Finally, explaining what such models learn has primarily been approached through feature attribution, with methods adapted from classification to search tasks [17, 18]. Other works examine learned representations directly [19, 20], with the geometry of the MoE-induced embedding spaces, however, remaining largely underexplored.

3. Methodology

Figure 1 illustrates the shared design. A bi-encoder backbone encodes queries and documents, with their embeddings then passed directly to the MoE layer. The MoE layer comprises n experts, each a feed-forward network with a down-projection layer that halves the input dimension, an up-projection

Table 1
Zero-shot evaluation comparing standard fine-tuning (FT), with SB-MoE (TOP-1, ALL), and DC3 (DENSEC3_W). All MoE-enhanced retrievers employ **6 experts**. Scores refer to nDCG@10. Symbol * indicates a statistically significant difference over FT. Statistical significance is verified through paired, two-sided Student's t-tests, applying the Bonferroni correction for multiple testing at $p = 0.05$ using the ASPIRE toolkit [25]. Best value per backbone/dataset/metric is **bold**.

Backbone	Variant	covid	nfc	nq	hotpot	fiqa	arguana	touche	quora	dbpedia	scidocs	fever	cfever	scifact	avg.
TinyBERT	FT	.338	.212	.245	.320	.129	.196	.131	.785	.223	**.102**	.447	**.149**	.365	.280
	SB_{TOP-1}	.348	.209	.250*	.320	.131	.201*	.142*	**.786**	.220	.101	.442*	.140*	.377*	.282
	SB_{ALL}	.356	.212	.250*	**.326***	.133*	.202*	.143*	**.786**	**.224**	.101	.452*	.138*	**.378***	.285
	DC3	**.447***	**.218**	**.253***	.315*	**.142**	**.219***	**.166**	.781*	.223	**.102**	**.453***	.129*	.371	**.294**
BERT-Small	FT	.305	.228	.210	**.445**	.141	**.371**	.049	.794	.221	**.119**	.314	.100	.424	.286
	SB_{TOP-1}	.303	.227	**.213***	.437*	.146*	.363*	.050	.788*	**.222**	.116*	.312*	.104*	.421	.285
	SB_{ALL}	.306	.229	.210	.440*	.147*	.363*	.047	**.795**	.219	.116*	.311*	.104*	.432	.286
	DC3	**.322**	**.250***	.198*	.442	**.165***	.340*	**.065***	.785*	.217	**.119**	**.422***	**.137***	**.502***	**.305**
BERT-Base	FT	**.489**	**.270**	**.330**	**.463**	**.221**	.273	**.146**	.824	**.290**	.127	**.605**	**.191**	.540	**.367**
	SB_{TOP-1}	.470	.265	.322*	.455*	.218	.281*	.143	**.825**	.279*	**.128**	.590*	.188*	.542	.362
	SB_{ALL}	.474	.266	.323*	.457*	.218	.283*	.144	**.825**	.277*	.127	.588*	.186*	**.543**	.362
	DC3	.483	**.270**	.308*	**.463**	.215	**.329***	**.146**	.824	.278*	.127	.568*	.167*	.511*	.361
Contriever	FT	.427	.333	.376	**.650**	.291	.302	.138	.864	.390	.166	.681	.185	**.679**	.422
	SB_{TOP-1}	.266*	.324*	.358*	.634*	.272*	.348*	.107*	.818*	.376*	.166	.614*	.197*	.665	.396
	SB_{ALL}	.265*	.326*	.363*	.638*	.273*	.350*	.116*	.818*	.377*	**.167**	.619*	.198*	.668	.398
	DC3	**.439**	**.338**	**.379**	.647	**.310***	**.369***	**.188***	**.865**	**.403***	.164	**.741***	**.213***	.666	**.440**
ColBERT	FT	.519	**.250**	.341	.430	.224	.274	.151	.807	.283	.111	.557	.158	.516	.355
	SB_{TOP-1}	.546*	.248	**.347***	.438*	.228	.267*	**.158**	.811*	**.285**	.110	.562*	**.160**	.516	**.360**
	SB_{ALL}	**.551***	.249	**.347***	.438*	**.231**	.266*	**.158**	.811*	.284	.110	.562*	**.160**	**.517**	**.360**
	DC3	.511	**.250**	.338	**.443***	.224	**.286***	.156	**.815***	.278	**.112**	**.594***	.148*	.497	.358

layer that restores it, and a skip connection [21]. Every expert produces a refined version of the input embedding, and a gating mechanism weights all experts' outputs. The pooling module aggregates the n outputs into a single representation used for similarity estimation, computed as their weighted sum $y = \sum_{i=1}^{n} f_i(x) \cdot g_i(x)$, where x is the input embedding, $f_i(x)$ the output of the i-th expert, and $g_i(x)$ its gating weight. Since MoE acts directly on the output embedding rather than inside the backbone, it adds only few parameters ($0.59M$–$3.85M$, from to +3.2% to +5.9% for 6 experts, depending on the backbone and gating mechanism), can be applied identically to queries and documents, and is agnostic to the underlying model. Hence, the two approaches differ only in how the gate is trained. In SB-MoE [1], the gating mechanism is a small feed-forward network trained jointly with the experts and the backbone in an unsupervised manner for the downstream IR task. Noisy Top-1 gating during training ensures that all experts are explored [16]. In DenseC3 [2], the gate is a multi-label classifier trained once offline on external data annotated with the six levels of Bloom's Taxonomy [22], which characterize texts by the Cognitive Complexity of their content [23]. It is then frozen and used to weight the experts, so that each expert specializes in one complexity level, and query and document representations of comparable conceptual difficulty are drawn into the same region within the embedding space. To examine how expert routing shapes the induced retrieval spaces, we propose ExpertLens [3], a post-hoc explainability framework for MoE-enhanced DRMs. ExpertLens leverages DeepView [24], which relies on discriminative dimensionality reduction to project high-dimensional points into two dimensions while preserving a classifier's decision boundaries, unlike conventional methods such as UMAP. ExpertLens uses a surrogate classifier distilled directly from the MoE gate, so that the visualized boundaries correspond to expert routing decisions.

4. Experimental Analysis

Each model is trained once on MSMARCO and evaluated without further adaptation on the test sets of 13 BEIR collections [26], following the protocol of the original works [1, 2]. The FT baseline is the same backbone fine-tuned on MSMARCO without any MoE layer, using identical training data and hyperparameters, so that the observed differences are attributable to expert routing (Table 1). Overall, both examined MoE-enhanced DRMs transfer well to unseen collections. For four of the five backbones, at least one expert routing strategy improves over FT on a majority of the 13 collections. The gains are unevenly distributed, and they concentrate on the collections that differ most from the training

distribution. Improvements on domain-specific or argumentative corpora are frequently substantial, reaching +32.25% on TREC-COVID with TinyBERT and +34.4% on FEVER with BERT-Small, both under the weakly supervised gating (DenseC3). On the Wikipedia-based collections closest to MSMARCO (e.g., NQ, HotpotQA), differences are small in either direction, suggesting that expert routing contributes most where the backbone alone generalizes least.

The two gating mechanisms do not benefit the same backbones. Unsupervised routing is strongest on TinyBERT and ColBERT, where it improves performance on 10 of the 13 collections, and it is the weaker option on Contriever. Weak supervision shows the opposite profile, as it proves to be the strongest variant on Contriever, improving on 10 collections with gains of up to 36.23% on Touché, while leading on BERT-Small as well. BERT-Base is the one backbone that neither strategy improves, with no variant exceeding FT on more than four collections. Since ColBERT and Contriever have the same parameter count as BERT-Base yet respond well to at least one expert routing strategy, this pattern does not follow model size. It instead separates BERT-Base, pre-trained with masked language modeling alone, from the distilled and retrieval-oriented backbones, indicating that the pre-training strategy strongly influences the MoE benefits to the downstream IR task. Visual and quantitative analyses of the MoE-induced embedding spaces [3] reveal the effect of expert routing. Both SB-MoE and DenseC3 place relevant documents *relatively* closer to their queries, while the *absolute* distance between them grows. Hence, expert routing expands the overall geometry of the embedding space while tightening the immediate query neighborhood. The expert subspaces also follow a consistent pattern. Under both gating mechanisms, a dominant expert absorbs most documents and stays close to the general corpus linguistic profile (e.g., lexical diversity, readability, etc.), while minority experts capture lexically richer text. Expert specialization therefore reflects the intrinsic linguistic properties of documents rather than their topic, and this holds regardless of whether the routing signal is learned or externally imposed.

5. Conclusions & Future Work

This extended abstract connects a line of work that investigates Mixture-of-Experts (MoE) approaches in dense retrieval tasks. The outcomes that emerge suggest that embedding contextualization through MoE is an inexpensive way to improve the generalization of dense retrievers, with its benefits surviving transfer to unseen tasks and domains. The observed performance gains are attributable to the selected expert routing strategy, with weakly supervised gating yielding the largest individual gains. These findings extend beyond the two models examined here, since the proposed MoE architecture operates directly on the output embedding of the backbone encoder. Therefore, it is agnostic to the underlying model and can be placed atop any architecture without requiring any alterations. Additionally, expert routing can be decided by an interchangeable gating mechanism. Replacing the gate with a differently trained classifier injects a distinct external signal into the representations. Cognitive Complexity is therefore one instantiation of a more general mechanism for enriching and contextualizing query and document embeddings with external information. Future work will explore alternative signals, such as domain, readability, or user expertise, and investigate which textual characteristics are most informative for expert routing and contribute most to retrieval effectiveness.

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Declaration on Generative AI

During the preparation of this work, the authors used Claude for grammar and spelling checks. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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